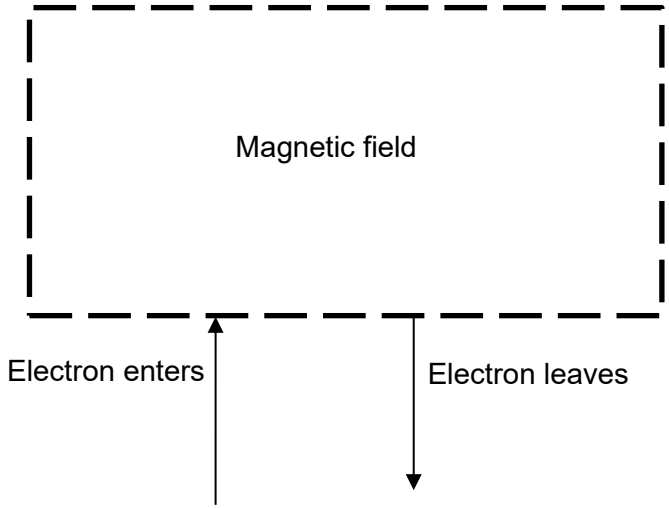
23 An electron travelling at a speed of 5.0×10^{7} m s^{-1} enters perpendicularly into a rectangular region of uniform magnetic field of strength 3.0 T, as shown below. The field is perpendicular to the page:



How much time does the electron spend inside the magnetic field?

A 2.3×10^{-31} s **B** 1.1×10^{-23} s
C 1.2×10^{-19} s **D** 6.0×10^{-12} s